Personalized Multiparty Few-Shot Learning for Remote Sensing Scene Classification

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***Abstract*— The existing few-shot scene classification (FSSC) algorithms have achieved satisfactory results, but they are limited by the paradigm of centralized machine learning, i.e., private remote sensing data need to be centralized on a certain server for training. However, remote sensing images generally contain sensitive information such as national security and company privacy, so it is realistically difficult to collect remote sensing data from all the parties. Therefore, there is a pressing requirement in FSSC for a novel paradigm to achieve multiparty collaborative learning without compromising remote sensing data privacy. In this article, we formulate a novel personalized multiparty few-shot learning (PMPFSL) paradigm for remote sensing scene classification. In PMPFSL, different participants can achieve multiparty collaborative learning without sacrificing the privacy of their local data, and their respective local models are able to recognize the unseen remote sensing scene categories with a small number of labeled samples. Importantly, the proposed PMPFSL is applicable to various multiparty learning (MPL) algorithms and FSSC networks. Moreover, to address the problems of local model overfitting and poor discriminability of few-shot metrics, we propose the personalized adaptive distillation (PAD) scheme and multiscale feature matching network (MSFMNet) on PMPFSL, respectively. Specifically, each participant obtains the MSFMNet with initialization parameters, and implements a certain number of local training on their respective private machines. Global aggregation is subsequently achieved by uploading only the local models to the central server. In a new round of local training, the participants realize personalized data adaptation to the global model based on the PAD. Overall, the proposed PMPFSL customizes a personalized few-shot model for each participant that is more tailored to their respective remote sensing scenarios. The experimental results demonstrate that our PMPFSL is superior in three benchmark FSSC datasets. We also extensively studied and analyzed the contributions of PAD and MSFMNet in the proposed PMPFSL framework.**

***Index Terms*— Few-shot learning, few-shot scene classification (FSSC), personalized multiparty learning (MPL), remote sensing data security.**

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I. Introduction

SCENE classification, which aims to assign predefined category labels to remote sensing images based on the functional and semantic information of the region or target contained in the scenario [1], [2], [3], has been widely used in various military and livelihood fields, such as environmental monitoring [4], urban planning [5], and disaster assessment [6]. In recent years, with the emergence of deep learning driven by large-scale labeled remote sensing data, massive scene classification approaches have achieved satisfactory performance [7].

Nonetheless, behind these successes is the support of large-scale annotation of remote sensing samples, which not only imposes a heavy burden of labeling costs, but also concedes the performance of recognizing unseen categories [8], [9], [10]. In response to these requirements, few-shot learning has come into the spotlight, which strives to parse task-relevant scenes in query images based on a small amount of labeled supporting data [11]. In the last few years, a large number of excellent few-shot remote sensing scene classification algorithms have emerged [12], [13], [14], [15], [16]. Approaches based on meta-learning training paradigm [17], [18], [19] are of particular interest. Ordinarily, episodic learning is used in the meta-learning phase, where the base dataset is constructed with multiple few-shot tasks for training. The aim is to enable the model to obtain the generalization ability across these category-changing meta-tasks [20], [21].

However, although the existing few-shot scene classification (FSSC) work has achieved encouraging results, the issue of remote sensing data security has been neglected [22], [23]. Data (including the base dataset for training and the novel dataset for testing) need to be stored and processed centrally, and there is a risk of privacy leakage [24]. In particular, remote sensing scene data is sensitive, which involves confidential national and geographical information, such as topography, resource layout, military equipment, and urbanization development [22], [25], [26]. Therefore, in the environment of increasingly strict data supervision, achieving

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multiparty collaborative learning under the guarantee of data privacy deserves to become a focus in the field of remote sensing.

To alleviate the above data privacy issues, some recent remote sensing-related work has resorted to multiparty learning (MPL) for exploration, such as object extraction of very-high-resolution (VHR) images [27], target recognition of multiradar data [28], and hyperspectral classification [29]. MPL, also known as federated learning, is an emerging technology that uses distributed learning and optimization methods to protect data privacy in multiparty collaborations [30], [31]. In MPL, participants (individuals, universities, remote sensing agencies or companies) only need to upload locally trained models to the central server without transmitting data, and obtain a global model with better generalization performance by aggregating model parameters [32].

Nevertheless, most of the current work related to MPL [27], [28], [29] only performs remote sensing task adaptation of federated aggregation algorithms and does not incorporate specific designs for the characteristics of remote sensing images. In reality, the remote sensing images owned by each party are heterogeneous, since their respective image categories and quantities generally do not satisfy the assumption of independent identical distribution. In addition, since the hardware facilities and computing power of each party are different, it is necessary to provide personalized solutions based on their respective data volume and model requirements. More importantly, most of the current MPL research is based on the traditional classification training paradigm, where the local model relies on a large number of labeled samples for updates, and do not have the ability to quickly generalize to unseen categories. Overall, data heterogeneity, lack of personalized solutions, and absence of the few-shot paradigm have emerged as current challenges of MPL in remote sensing scene classification.

To alleviate the abovementioned challenges, in this article, we propose a novel personalized multiparty few-shot learning (PMPFSL) framework. Based on the PMPFSL, we specifically designed for the remote sensing scene classification task in personalized adaptive distillation (PAD) scheme and few-shot network architecture. The contributions of our work are summarized below.

1) A general PMPFSL paradigm is formulated, which is applicable to any personalized MPL strategy and few-shot remote sensing scene classification network. The proposed PMPFSL provides a novel multiparty collaborative training paradigm, which enables multiple participants to obtain personalized few-shot models that are more compatible with local data with only a small number of labeled samples, without involving their sensitive remote sensing scene data. To the best of our knowledge, there are still no relevant studies.
2) A PAD scheme is designed to combine the local and global models for adaptive weight initialization and prevent local model overfitting in training. The PAD achieves local model initialization by retaining the shallow layers of the local models, while personalizing the deeper layers with adaptive aggregation weights according to the characteristics of local remote sensing data. Meanwhile, to overcome the overfitting problem of local models in few-shot settings, we perform the Lipschitz continuity guided-based distillation regularization strategy for training.
3) A multiscale feature matching network (MSFMNet) is proposed to construct an internal self-attention matching metric specifically designed for remote sensing scenarios. Existing metric-based FSSC methods [13], [15], [21], [33] are targeted at the last layer of representation extracted from the backbone for matching the support features and query features. However, in complex remote sensing scenarios, the problems of large intra-class variability and high inter-class similarity are serious, especially in the case of a small number of labeled samples. Therefore, we investigate a self-attention matching metric with multiple scales to construct intrinsic associations for different sophisticated targets.

The remainder of this article is constructed as follows. Section II briefly reviews the related work on MPL and few-shot remote sensing scene classification. In Section III, our proposed PMPFSL framework and the specific architecture and components are described in detail. Section IV reports and analyzes the comparison and the ablation experimental results. Conclusions are drawn in Section V.

II. Related Work

A. Multiparty Learning

MPL, also known as federated learning, is a distributed machine learning approach aimed at training and learning on local devices, keeping the model training process on users' terminal devices instead of transferring the raw data from the dataset to a central server [34]. In MPL, multiple participants, such as mobile or edge devices, collaborate to train a global model without sharing their raw data. Instead, each participant only shares locally processed model parameter updates [35]. In simple terms, MPL achieves data privacy and security by performing training and parameter updates on local devices, providing a solution for collaboration among distributed devices, and facilitating the construction of more privacy-preserving and secure machine learning models.

McMahan et al. [30] first proposed the paradigm of MPL, known as FedAvg, where each participant trains their own model on local datasets and then sends the updated model parameters to a central server for average aggregation. To alleviate the difficulty of heterogeneity between clients, Tan et al. [32] proposed the FedProto to aggregate the local prototypes collected from different clients, and sent the global prototype back to all clients to standardize the training of local models. While the FedProx [31] introduced the regularization term on the basis of FedAvg [30] to solve the heterogeneity in MPL.

Personalized MPL is specifically explored in the direction of poor convergence on highly heterogeneous data, and lack of personalization of models for local tasks or datasets. Fallah et al. [36] combined the meta-learning with FedAvg, called PerFedAvg, obtained the initial model based on the

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local data of the client, and then performed several gradient descents on the initial model locally to obtain their own personalized models. Dinh et al. [37] introduced the Moreau Envelopes in the client loss function based on PerFedAvg to obtain the personalized local model. Instead of computing individual model averages with constant weights for the entire federation, Zhang et al. [38] proposed the FedFomo, which efficiently computes the individualized combinations of optimally weighted models for each client based on the extent clients benefit from another model. In [39], the concept of inHerited Private Model was introduced to supervise the next round of global aggregation with a local model of historical personalization.

However, in the field of remote sensing, research on MPL is still in its infancy. In [22], a review on artificial intelligence security in remote sensing summarizes five important aspects, including federated learning, adversarial attacks, backdoor attacks, uncertainty, and explainability, and provides researchers with available code and datasets to move this promising field forward. In specific studies, Zhang et al. [27] designed a federated prototype matching framework for VHR remote sensing objective detection, and combined with a sparse ternary compression strategy to reduce the communication cost. Song and Dong [28] proposed a distributed MPL approach for moving and stationary target capture and recognition in SAR images to alleviate the sample imbalance problem across clients. Zhu et al. [40] presented a blockchain-enabled remote sensing image classification framework, aiming to defend the encryption models from malicious attacks without compromising user privacy data. Cai et al. [29] applied MPL to hyperspectral classification and designed a stereo attention cross-decoupling fusion to extract more discriminative features. In [41], a multiparty personalized collaborative learning framework is proposed for dealing with the privacy and sensitivity issues of remote sensing data by decoupling personalized model optimization and global model learning in communication iterations, allowing different participants to co-train and share the models while preserving the privacy of the local data.

B. Few-Shot Remote Sensing Scene Classification

According to the training paradigm, few-shot remote sensing scene classification is currently based on meta-learning and transfer learning [42]. Since our study belongs to the former, i.e., episodic training, we focus on introducing and analyzing the meta-learning-based few-shot remote sensing scene classification work below. Specifically, the meta-learning few-shot paradigm based on episodic training can be further divided into optimization-based [12] and metric-based [13], [15], [17]. The optimization-based approaches accomplish fitting in the few-shot scenario by adapting the training optimization strategy, while the metric-based approaches explore the distance measure between the support set and the query set in the feature dimension [43].

Zhai et al. [12] constructed a lifelong scene classification (LLSR) framework that generalizes the model to novel datasets by optimizing iterations with a small number of samples. Li et al. [13] proposed discriminative learning for adaptive matching network (DLA-MatchNet) to enhance the discriminability of metric features by incorporating the channel attention mechanism. In [14], graph-based embedding smoothing network (GES-Net) was designed to incorporate unsupervised nonparametric regularizers to obtain better-embedded features and to represent the relationships between features at the task level by constructing graphs. Huang et al. [44] presented a task-adaptive embedding network (TAE-Net) to obtain better feature representation by integrating pre-training and task-adaptive attention mechanisms. Cheng et al. [15] proposed a Siamese-prototype network (SPNet), which corrects the deviation of the feature prototype by designing a novel loss function with fusing self- and inter-calibration. In [16], the task-specific contrastive learning (TSC) framework combined with the self-attention mechanism was designed to obtain more discriminative feature embeddings. In [17], a scene-graph-matching-based meta-learning called SGMNet explores the spatial correlation of different objectives in a scene with graph construction. Yuan et al. [20] proposed the metric learning and local descriptors (MLLD) to smooth the decision boundary thereby increasing the low confidence space of the network. In [33], a multiscale covariance network (MCMNet) explored the similarity of prototypes in the feature extractor for different scales of manifold space. Chen et al. [21] designed a multimanifold metric learning (MMML) to embed two heterogeneous manifold structures for feature fusion by introducing the Riemann kernel function.

Overall, the existing few-shot remote sensing classification methods are trained based on the centralized machine learning paradigm, which means that both the data and the model need to be centralized on one machine. This paradigm easily leads to data leakage. More importantly, due to the sensitivity and confidentiality of remote sensing data, it is difficult to collect data from multiple remote sensing institutions for model training in real applications. Therefore, to realize the generalization of the model with a small number of samples while protecting the privacy of the data is a direction worthy of attention and exploration in the remote sensing field.

III. Methodology

A. Problem Statement

We formulate a framework for generalizing the model to unseen classes with only a small number of annotated remote sensing images for reference in the framework of MPL. In the few-shot setting, the dataset is divided into a base dataset $\mathcal{D}_{\text{base}}$ for training and a novel dataset $\mathcal{D}_{\text{novel}}$ for testing, with no overlapping scene categories between them. Assuming that there are a total of I participants to perform MPL, where each participant is represented as c_i. We adopt the paradigm of episodic training [45], [46], where in the N-way K-shot setup, we first randomly select N categories from the dataset, and then select K labeled images inside these N categories to construct the support set $\mathcal{S}$. While the query set $\mathcal{Q}$ consists of unlabeled images selected from one of these N categories. Based on this, the privacy data of each participant is represented as $\mathcal{D}_i = \{(\mathcal{S}_m^i, \mathcal{Q}_m^i)\}_m^M$, where M is the number

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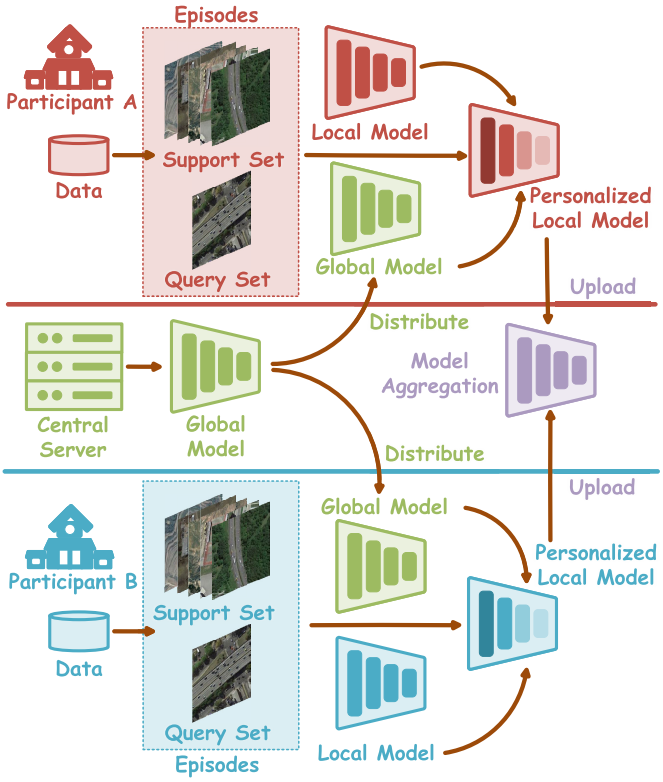


Fig. 1. Schematic of the global communication steps for a round of two participants in PMPFSL framework.

of episodes. $\mathcal{D}_i$ is constructed in $\mathcal{D}_{\text{base}}$ and $\mathcal{D}_{\text{novel}}$ in training and testing phase, respectively.

In MPL, participants update their models locally and then upload the parameters to a central server for model aggregation, the aggregated model is then distributed to the participants, which is defined as a round of global communication. In the tth round of global communication, the local model parameter for participant c_i is represented as θ_i^t, while the aggregated global model parameters can be expressed as φ^t. For each participant, the loss function can be indicated as $\mathcal{L}_i = \mathcal{L}(\theta_i; \mathcal{D}_i)$. Based on the loss function $\mathcal{L}_i$, participant c_i updates their model parameters θ_i based on the local data $\mathcal{D}_i$. Without loss of generality, there are large differences in the distribution of data for each participant in terms of category or quantity. The purpose of MPL is to obtain a generalized global model φ^* from these local models that exist in silos. While in our personalized setting, we simultaneously pursue the optimal set of personalized model parameters $\{\theta_i^*\}_i^I = \arg\min_{\{\theta_i\}_i^I} \sum_{i=1}^I \mathcal{L}_i$. Based on the above settings, the program should satisfy the following properties.

1) Data Privacy: In MPL, participants' private data does not need to leave the local device but is computed and model trained locally, avoiding leakage and misuse of sensitive data. Each participant does not have permission to access the private data of the other participants, and it is ensured that the other party's data information and the distribution of raw data cannot be deduced. In terms of data privacy, PMPFSL can be equipped with various privacy-preserving strategies such as differential privacy [47], homomorphic encryption [48], and functional encryption [49].

Algorithm 1 PAD

Input: I: the number of participants; T: the number of global communication rounds; E: the number of local updated episodes.

Output: Optimal personalized models $\{\theta_1^*, \ldots, \theta_N^*\}$.

```
Initialize the global model φ^0.
for t = 1 to T do
    Server sends the global model φ^{t-1} to the participants.
    for i = 1 to I do
        Update the adaptive aggregating weights W_i by Formula (2).
        Participant c_i obtains the local initialization model θ^t according to Formula (1).
        Train the local model θ^t based on Formula (3) with E episodes.
    end for
    Participants send the updated local model θ^t to the server.
    Update the global model φ^t ← (1/I) Σ_{i=1}^{I} θ_i^t.
    t ← t + 1.
end for
```

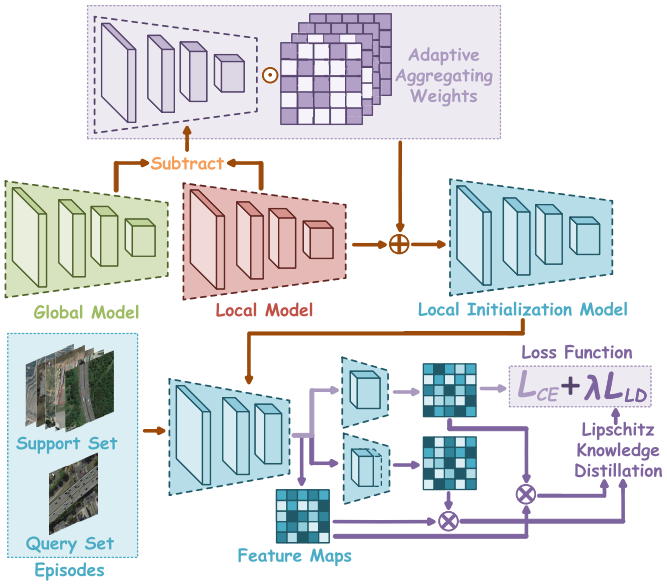


Fig. 2. Diagram of the PAD scheme of one participant.

2) Improved Performance: The aggregated global model cannot be sufficiently less predictive of local samples than locally trained models, otherwise, participants are not motivated to engage in collaboration. In addition, for locally unseen samples, the generalization performance of the distributed global model should be better than that did not participate in MPL.

3) Personalized Solutions: In reality, the distribution of participants' private data is usually not satisfied with being independently and identically distributed (IID). Therefore, it is necessary to provide personalized models based on the data characteristics of different participants, rather than directly distributing a simple aggregated global model.

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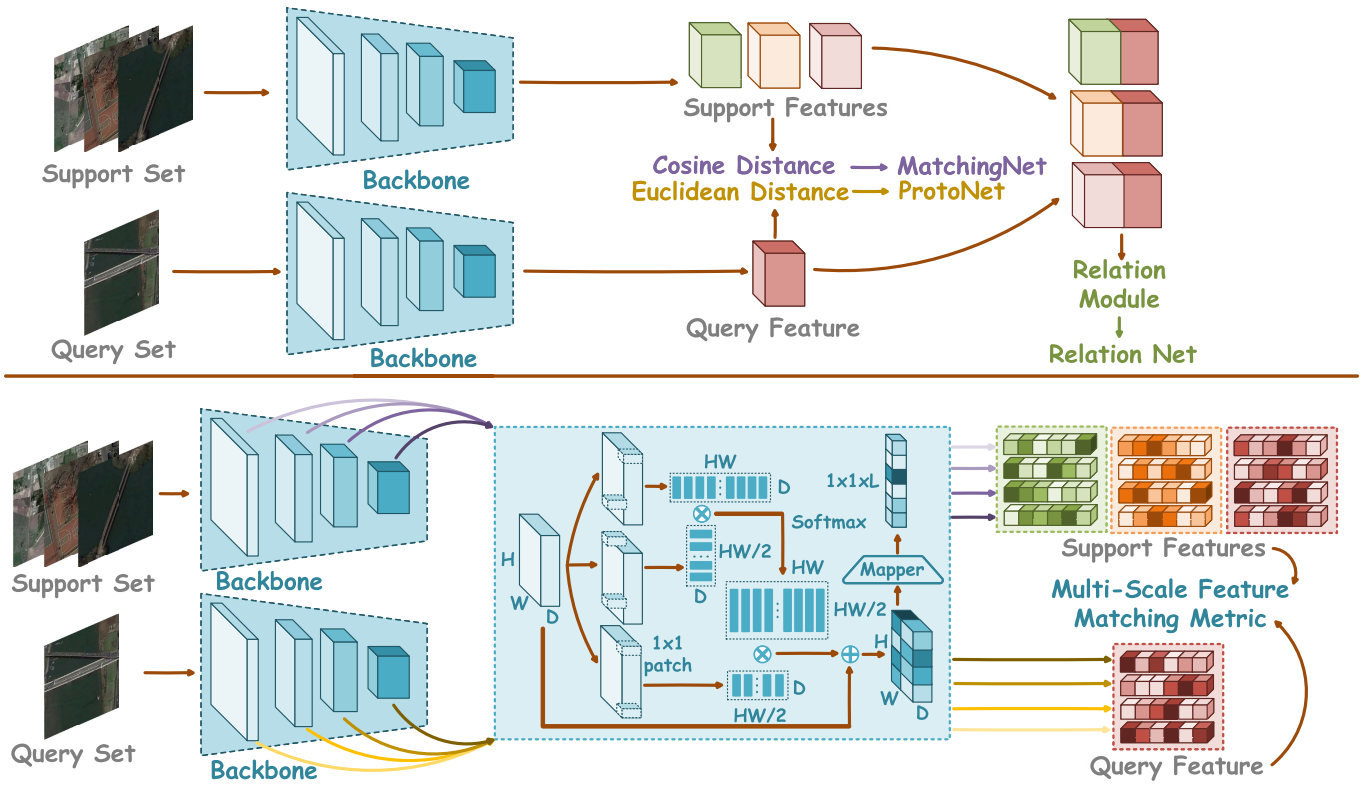


Fig. 3. Differences between the final layer of features and multiscale feature metrics. We show the case of a three-way one-shot in the figure. (Top) MatchingNet [45] with cosine distance, ProtoNet [46] with Euclidean distance, and Relation Net [50] with relation module as the metrics. (Bottom) Proposed MSFMNet. In MSFMNet, the features corresponding to the four scales of each block in the backbone are input to the self-attention module and unified with different mappers to obtain a $1 \times 1 \times L$ metric features.

B. Personalized Multiparty Few-Shot Learning Framework

The training paradigm followed by most previous MPL frameworks is to divide the dataset into a training set, a validation set, and a test set, which all have the same categories. Heterogeneous data, on the other hand, is studied by assigning samples of different categories to the participants. In this framework, the local model updating is actually still dependent on a large number of labeled samples, and the generalizability of the local models to unseen categories is achieved by simple global model aggregation. In this article, we formulate a novel PMPFSL framework, which can piggyback on various MPL algorithms and FSSC architectures.

In PMPFSL, participants can voluntarily engage in multiparty collaborative learning through multiple global communication rounds, to obtain an FSSC network with better generalization without compromising local private data. To more clearly articulate the specific processes, we show a simple case in Fig. 1 with a schematic of the global communication steps for a round of two participants in the PMPFSL framework. First, the central server distributes the global model to the participants who responded to the communication in the current global round. Subsequently, participants aggregated the global and local models into a personalized initialization model with the personalization strategy. Participants trained their respective models locally based on their computing power and data volume. It is worth noting that the training and test set categories of each participant are not intersected, but the dataset is divided into multiple episodes. The local models learn to quickly recognize unseen remote sensing scene categories with only a small number of labeled samples from these episodes. Afterward, participants uploaded the trained local models to the centralized server. With a certain number of global communications, participants can obtain a personalized model that is more generalized but also performs better on local data.

The superiority of PMPFSL over previous MPL algorithms and personalized MPL algorithms lies in the few-shot learning scheme and specific personalized strategies for remote sensing scene classification. Previous MPL and personalized MPL algorithms divide the participant's dataset into training, validation, and test sets, and the categories inside these sets are identical. Heterogeneous data is studied instead by assigning different categories to the participants, but the categories for training and testing are still the same for each participant. While in the PMPFSL, each participant's data are divided into the base and novel datasets, which do not have overlapping categories. In this way, the functionality of the model changed from identifying only previously seen categories to recognizing novel categories with only a small number of labeled references. Moreover, the combination of specific personalization strategies allowed participants to obtain a few-shot architecture with better generalization performance and more suitable for local remote sensing scene categories at the same computational cost without disclosing local data.

C. Personalized Adaptive Distillation Scheme

In general MPL [30], participants directly perform a new round of local model updates based on the global model

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TABLE I
OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT REMOTE SENSING SCENE CLASSIFICATION METHODS ON THE WHU-RS19 DATASET

Method	5-way	
	1-shot	5-shot
Centralized Machine Learning		
MatchingNet [45]	60.14 ± 0.67	68.66 ± 0.38
MAML [55]	49.13 ± 0.65	62.49 ± 0.51
ProtoNet [46]	69.03 ± 0.65	84.84 ± 0.33
TPN† [56]	66.51 ± 0.87	78.50 ± 0.56
Relation Net [50]	70.01 ± 0.68	86.04 ± 0.32
Meta-SGD [57]	51.54 ± 2.31	61.74 ± 2.02
LLSR [12]	57.10	70.65
DLA-MatchNet [13]	68.27 ± 1.83	79.89 ± 0.33
IDLN [58]	73.89 ± 0.88	83.12 ± 0.56
PCFGNet [59]	72.41 ± 0.91	85.26 ± 0.66
GES-Net† [14]	75.84 ± 0.78	82.37 ± 0.38
TAE-Net [44]	73.67 ± 0.74	$\underline{88.95 \pm 0.52}$
SPNet [15]	$\underline{81.06 \pm 0.60}$	88.04 ± 0.28
TSC [16]	70.99 ± 0.74	82.18 ± 0.32
SGMNet [17]	$\mathbf{85.06 \pm 0.55}$	90.12 ± 0.27
MLLD [20]	76.07 ± 0.65	$\mathbf{90.69 \pm 0.27}$
MLCMNet [33]	72.73 ± 0.48	84.54 ± 0.23
MMML [21]	77.76 ± 0.69	88.40 ± 0.54
MSFMNet	78.24 ± 0.42	88.16 ± 0.58
Multi-Party Learning		
PMPFSL$_{\text{[ProtoNet]}}$	65.21 ± 0.90	81.37 ± 0.56
PMPFSL$_{\text{[RelationNet]}}$	66.56 ± 0.61	82.53 ± 0.59
PMPFSL$_{\text{[MatchingNet]}}$	58.14 ± 0.66	67.19 ± 0.45
PMPFSL$_{\text{[MSFMNet]}}$	$\mathbf{77.24 \pm 0.32}$	$\mathbf{88.66 \pm 0.37}$

TABLE II
OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT REMOTE SENSING SCENE CLASSIFICATION METHODS ON THE UC MERCED DATASET

Method	5-way	
	1-shot	5-shot
Centralized Machine Learning		
MatchingNet [45]	46.46 ± 0.75	67.39 ± 0.50
MAML [55]	48.86 ± 0.74	60.78 ± 0.62
Meta-SGD [57]	50.52 ± 2.61	60.82 ± 2.00
ProtoNet [46]	54.03 ± 0.78	70.81 ± 0.48
TPN† [56]	53.36 ± 0.77	68.23 ± 0.52
Relation Net [50]	49.77 ± 0.78	57.98 ± 0.57
LLSR [12]	39.47	57.40
RS-MetaNet [60]	57.23 ± 0.56	76.08 ± 0.28
DLA-MatchNet [13]	53.76 ± 0.62	63.01 ± 0.51
MKN [61]	58.45 ± 0.54	77.92 ± 0.32
SCL-MLNet [62]	51.37 ± 0.79	68.09 ± 0.92
GES-Net† [14]	58.88 ± 0.81	$\mathbf{81.66 \pm 0.50}$
TAE-Net [44]	60.21 ± 0.72	77.44 ± 0.51
SPNet [15]	57.64 ± 0.73	73.52 ± 0.51
TSC [16]	55.11 ± 0.68	69.20 ± 0.64
SGMNet [17]	60.52 ± 0.74	73.42 ± 0.49
MLLD [20]	59.05 ± 0.75	77.76 ± 0.52
MLCMNet [33]	56.58 ± 0.39	71.35 ± 0.46
MMML [21]	$\underline{61.42 \pm 0.34}$	79.95 ± 0.16
MSFMNet	$\mathbf{62.85 \pm 0.32}$	$\underline{80.88 \pm 0.35}$
Multi-Party Learning		
PMPFSL$_{\text{[ProtoNet]}}$	51.06 ± 0.62	68.82 ± 0.47
PMPFSL$_{\text{[RelationNet]}}$	47.64 ± 0.37	55.72 ± 0.58
PMPFSL$_{\text{[MatchingNet]}}$	44.28 ± 0.47	65.45 ± 0.62
PMPFSL$_{\text{[MSFMNet]}}$	$\mathbf{64.08 \pm 0.35}$	$\mathbf{81.69 \pm 0.42}$

received from the central server. However, the global model, which aggregates the weight parameters of each participant, is not necessarily valid for every participant, especially for the realistic case of heterogeneous remote sensing data. Therefore, it is necessary to personalize the local model according to the characteristics of each participant's local data. Here, we design a PAD scheme exclusive for few-shot remote sensing scene classification, as shown in Algorithm 1.

We illustrate more graphically the process of model fusion and training of participant's personalized model in a global communication round with Fig. 2. In traditional MPL, after receiving the global model, participants directly initialize the local model with the received global model, i.e., $\{\theta_i^t\}_{i=1}^I \leftarrow \varphi^{t-1}$. Instead, for each participant c_i, we implement local model initialization by aggregating global and local models for adaptive aggregation as follows:

$$\theta_i^t \leftarrow \theta_i^{t-1} + \left(\varphi^{t-1} - \theta_i^{t-1}\right) \odot W_i \tag{1}$$

where "$\odot$" is the Hadamard product, and W_i represents the adaptive aggregating weights for participant c_i, defined as follows:

$$W_i^l = \begin{cases} \mathbf{1}, & l \le \zeta \\ W_i^l - \eta_w \nabla \mathcal{L}_{W_i^l}\left(\theta_i^t, \mathcal{D}_i; \theta_i^{t-1}\right), & l > \zeta \end{cases} \tag{2}$$

where $\zeta \in [1, |\theta_i|]$ is a parameter that controls the number of adaptive layers, $|\theta_i|$ is the number of layers of local model, and η_w denotes the learning rate of parameter learning. For each element in W_i, the weight clipping is employed, i.e., $\forall w \in W_i$, $w \in [0, 1]$, $\sigma(w) = \max(0, \min(1, w))$.

Specifically, in order to accurately capture the personalized information required by the participant from the global model, the shallow weights of local model are set to be consistent with the received global model, considering that the shallow layers are more capable of learning generalized information compared to the deep layers. As for the relatively deep weights of the local model, we achieve adaptive adaptation by collecting a certain amount of local data for learning.

After obtaining the local initialization model, the traditional MPL algorithms upload the model to the central server after a certain number of rounds of training locally. However, in the few-shot learning setting, since the test samples are the classes that have not been seen in training, this makes the local model prone to overfitting especially in heterogeneous data, which also leads to the aggregated global model not being able to obtain well-generalized capability. Therefore, we alleviate the above problem and simultaneously personalize each participant's local model by introducing a regularization term for distillation.

Particularly, as mentioned earlier, the shallow layers of local models are generalized for the distributed training paradigm, whereas deep layers are more oriented toward localized data features. Therefore, we perform Lipschitz continuity guided-based distillation regularization on the deep layers in

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Fig. 4. Some failure cases from MSFMNet on three benchmark few-shot datasets.

TABLE III
OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT REMOTE SENSING SCENE CLASSIFICATION METHODS ON THE NWPU-RESISC45 DATASET

Method	5-way	
	1-shot	5-shot
Centralized Machine Learning		
MatchingNet [45]	45.80 ± 0.86	64.35 ± 0.65
MAML [55]	48.40 ± 0.82	62.90 ± 0.69
Meta-SGD [57]	60.63 ± 0.90	75.75 ± 0.65
ProtoNet [46]	62.43 ± 0.84	78.68 ± 0.55
TPN† [56]	66.51 ± 0.87	78.50 ± 0.56
Relation Net [50]	66.58 ± 0.91	76.11 ± 0.58
RS-MetaNet [60]	52.78 ± 0.09	71.49 ± 0.81
LLSR [12]	51.43	72.90
DLA-MatchNet [13]	68.80 ± 0.70	81.63 ± 0.46
IDLN [58]	75.25 ± 0.75	84.67 ± 0.23
PCFGNet [59]	72.05 ± 0.75	85.07 ± 0.45
GES-Net† [14]	70.83 ± 0.85	82.27 ± 0.55
MKN [61]	65.84 ± 0.89	82.67 ± 0.55
SCL-MLNet [62]	62.21 ± 1.12	80.86 ± 0.76
TAE-Net [44]	69.13 ± 0.83	82.37 ± 0.52
SPNet [15]	67.84 ± 0.87	83.94 ± 0.50
TSC [16]	73.26 ± 0.15	84.62 ± 0.35
SGMNet [17]	70.40 ± 0.83	82.32 ± 0.53
MLLD [20]	65.88 ± 0.83	82.06 ± 0.53
MLCMNet [33]	74.61 ± 0.18	86.57 ± 0.42
MMML [21]	$\mathbf{77.35 \pm 0.73}$	88.19 ± 0.47
MSFMNet	72.03 ± 0.45	$\mathbf{88.62 \pm 0.46}$
Multi-Party Learning		
PMPFSL$_{[\text{ProtoNet}]}$	60.63 ± 0.51	76.56 ± 0.29
PMPFSL$_{[\text{RelationNet}]}$	64.29 ± 0.58	74.75 ± 0.53
PMPFSL$_{[\text{MatchingNet}]}$	42.96 ± 0.58	62.16 ± 0.25
PMPFSL$_{[\text{MSFMNet}]}$	$\mathbf{72.67 \pm 0.19}$	$\mathbf{89.54 \pm 0.17}$

the last block of local model, with the following loss function definition:

$$\mathcal{L}_i = \mathcal{L}_{\text{CE}}(\theta_i; \mathcal{D}_i) + \eta \mathcal{L}_{\text{LD}}\left(\left\|\Psi_i^T\right\|_{\text{SN}}, \left\|\Psi_i^S\right\|_{\text{SN}}\right) \quad (3)$$

where $\|\Psi_T\|_{\text{SN}} = (\xi_{\theta_{B-1}})^T \xi_{\theta_B}$, $\|\Psi_S\|_{\text{SN}} = (\xi_{\theta_{B-1}})^T \xi_{\theta_B^\omega}$, $\|\cdot\|_{\text{SN}}$ is the spectral norm, $\xi_{\theta_{B-1}}$ and ξ_{θ_B} are the feature maps output by the preceding block and the last block, respectively. $\xi_{\theta_B^\omega}$ denotes the feature map output by last block with the ω width reduction.

The loss function aims to promote the local models to learn generalized representations based on their own data by the regularization term, rather than forcing local models to approach the global model. After local training is completed, participants upload the local models to the central server for global aggregation and perform the next round of global communication.

D. Multiscale Feature Matching Network

Most previous FSSC algorithms [13], [15], [19], [33], [45], [46], [50] are based on the last layer feature out of the network to measure the support set and query set. In particular, the most representative of these as benchmark models are MatchingNet [45] with cosine distance, ProtoNet [46] with Euclidean distance, and Relation Net [50] with relation module as the metrics, as illustrated in Fig. 3 (top). However, although the deep features in the network possess better semantic information, their low-resolution results in poor perception of details. While the complex remote sensing scenes require fine-grained information of multiple objectives for discrimination. In the case of a small number of labeled samples, relying only on the global information of deep features is apparently not appropriate for scene classification.

Taking this line, we propose a novel MSFMNet as the metric, as shown in Fig. 3 (bottom). First, we are required to extract the set of features $\{\mathcal{F}_b\}_{b=1}^B$ output by each block in the feature extractor $f(\theta)$. Generally, the common feature extractors for few-shot learning such as Conv-4 or ResNet-12 have a total of four blocks. Subsequently, these features are input into the self-attention mapper separately to capture

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the dependencies of different targets in the scene. For the bth feature, following [51], the self-attention feature can be obtained as follows:

$$\mathcal{F}_b^{\alpha} = \text{Softmax}\left(\frac{Q\left(\mathcal{F}_b|\theta_b^q\right)K\left(\mathcal{F}_b|\theta_b^k\right)^T}{\sqrt{d_k}}\right)V\left(\mathcal{F}_b|\theta_b^v\right) \quad (4)$$

where $Q(\cdot)$, $K(\cdot)$, and $V(\cdot)$ are composed of 1×1 non-overlapping patches from $\mathcal{F}_b$, and $\sqrt{d_k}$ is the scaling factor.

Considering that the self-attention module has been widely used and is not our innovation, the specific procedures are not described in detail. Here, the strategy of fusing multiscale self-attention features and making predictions is not adopted, because it is still a prototype-based comparison without feature-level exploration of the correlation between the support set and the query set. Therefore, we design a Mapper consisting of a convolutional layer and an average pooling layer to map these features of different scales into the vectors with the same dimension ($1 \times 1 \times L$) as follows:

$$\mathcal{P}_b \in \mathbb{R}^{1\times 1\times L} \leftarrow \text{Mapper}_b\left(\mathcal{F}_b^{\alpha}\right). \quad (5)$$

It is worth noting that even for the same input features, $\mathcal{P}_b$ of the mapper is not unique. This is determined by the incomplete overlapping sampling of the patches for the self-attention input, so it is possible to map several different vectors simultaneously for features of a certain scale with the multihead self-attention attention module. Here, we employ $\mathcal{P}_b'$ to denote the set of vectors $\mathcal{P}_b$ extracted in bth blocks.

Overall, for a remote sensing scene image, MSFMNet extracts a collection of features rather than a feature of a particular layer or a multiscale fusion. Naturally, for a pair of support set and query set $(\mathcal{S}, \mathcal{Q})$, its output feature set pair is $(\mathcal{P}'^{\mathcal{S}}, \mathcal{P}'^{\mathcal{Q}})$, where $\mathcal{P}' = \{\mathcal{P}_b'\}_{b=1}^{B}$.

In an N-way K-shot few-shot problem, the support set has $N \times K$ images corresponding to $N \times K$ feature sets. For the nth category in the support set, we first obtain the respective barycenter as follows:

$$\widetilde{\mathcal{P}}'^{\mathcal{S}_n} = \frac{1}{|\mathcal{S}_n|}\sum_{k=1}^{K}\mathcal{P}'^{\mathcal{S}_{n,k}} \quad (6)$$

where $\mathcal{S}_{n,k}$ is the kth sample in support set of nth category.

Subsequently, the distance between the feature set of the query sample $\mathcal{Q}$ and each barycenter in the support set can be computed as

$$\Theta\left(\mathcal{P}'^{\mathcal{Q}}, \widetilde{\mathcal{P}}'^{\mathcal{S}_n}\right) = \sum_{p=1}^{|\mathcal{P}'|}\min_{q=1}^{|\mathcal{P}'|}\left\{-\cos\left(\mathcal{P}_p'^{\mathcal{Q}}, \widetilde{\mathcal{P}}_q'^{\mathcal{S}_n}\right)\right\}. \quad (7)$$

In the inference stage, MSFMNet aims to make the query sample $\mathcal{Q}$ to be found belonging to the $\mathcal{N}$th category in the referable support set, where $n = \{1, \ldots, N\}$, which can be expressed as

$$\Gamma(y = \mathcal{N}|\mathcal{Q}, \mathcal{S}) = \frac{\exp\left(-\Theta\left(\mathcal{P}'^{\mathcal{Q}}, \widetilde{\mathcal{P}}'^{\mathcal{S}_{\mathcal{N}}}\right)\right)}{\sum_{\mathcal{S}^n\in\mathcal{S}}\exp\left(-\Theta\left(\mathcal{P}'^{\mathcal{Q}}, \widetilde{\mathcal{P}}'^{\mathcal{S}_n}\right)\right)}. \quad (8)$$

IV. Experimental Results

A. Datasets

According to the paradigm of few-shot classification, the dataset is divided into the training set, validation set, and test set. In our experiments, the partitioning of these three benchmark datasets is consistent with most studies on FSSC [11], [13], [14], [15], [16], [17], [42], [44].

1) WHU-RS19 Dataset [52]: The WHU-RS19 is a satellite image dataset containing 1005 images with a resolution of 0.5 m, released by Wuhan University in 2012. These images are 600×600 pixels in size and cover 19 different scene categories, of which there are nine, five, and five classes in the training, validation, and test sets, respectively. The WHU-RS19 dataset has a large scale for each image although the categories are fewer, which puts more stringent requirements on the model to recognize multiple targets mixed together in a scene.

2) UC Merced Dataset [53]: The UC Merced dataset was released in 2010 by the University of California, Merced, which is used to classify land use scenes in urban areas. This dataset contains 21 categories of remote sensing scenes, with 100 images in each category, totaling 2100 images. The images are all extracted from the USGS National Map, with a spatial resolution of 0.3 m and an image scale of 256×256 pixels. The categories of the training set, validation set, and test set are 10, 6, and 5, respectively. The UC Merced dataset is rich in scenario categories and concentrated in urban and suburban scenarios, which puts demands on the diversity and stability of the few-shot model.

3) NWPU-RESISC45 Dataset [2]: The NWPU-RESISC45 dataset was created by Northwestern Polytechnical University in 2017. This dataset contains 31 500 images covering 45 scene classes with 700 images per class. These images have a resolution of 0.2–30 m and a size of 256×256 pixels. The categories of the training set, validation set, and test set are 25, 10, and 10, respectively. The scale of NWPU-RESISC45 is large and the complexity scenarios are complete, which can effectively test the few-shot model for solving the problem of large inter-class similarity and high intra-class diversity.

B. Implementation Details

In the experiments, we implemented all methods on a dual-card NVIDIA GeForce RTX 4090 with CUDA version 12.0. We report the overall accuracy using the most common settings of five-way one-shot and five-way five-shot with a confidence interval of 95%. Following the setting of the majority of previous few-shot studies, there are 15 query samples per episode. The backbone of MSFMNet is ResNet-12. For all datasets, the input image size is cropped to 84×84 pixels. The initial value of the learning rate is set to $5e^{-2}$, the weight decay parameter of the adaptive moment estimation (ADAM) optimizer is $5e^{-3}$, and the scaling is 0.5 every 20 epochs.

In terms of the parameters for MPL, we follow the settings in the widely recognized work [30], [31], [32], [36], [37], [38], [39]. Specifically, the number of global communication rounds and the number of local updated epochs are set to 200 and 300,

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TABLE IV
COMPARATIVE PERFORMANCE RESULTS (IN %) BETWEEN PMPFSL AND MPL ALGORITHMS ON THREE BENCHMARK DATASETS FOR THE FOUR BASELINE NETWORKS IN THE FEW-SHOT SETTING

Method	*baseline*[ProtoNet]		*baseline*[RelationNet]		*baseline*[MatchingNet]		*baseline*[MSFMNet]	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
WHU-RS19 Dataset								
FedAvg [30]	61.54 ± 0.47	75.71 ± 0.50	62.07 ± 0.35	77.11 ± 0.88	52.83 ± 0.42	62.89 ± 0.19	71.90 ± 0.34	82.94 ± 0.61
FedProx [31]	62.20 ± 0.53	76.47 ± 0.89	62.66 ± 0.47	77.58 ± 0.62	53.69 ± 0.56	63.43 ± 0.39	72.63 ± 0.28	83.86 ± 0.66
FedProto [32]	62.63 ± 0.73	77.33 ± 0.19	63.23 ± 0.31	78.49 ± 0.36	54.15 ± 0.28	64.11 ± 0.62	73.33 ± 0.21	84.76 ± 0.35
PerFedAvg [36]	63.07 ± 0.47	78.01 ± 0.70	63.75 ± 0.83	79.36 ± 0.26	54.87 ± 0.47	64.52 ± 0.44	74.00 ± 0.43	85.23 ± 0.32
pFedMe [37]	63.54 ± 0.63	78.99 ± 0.10	64.21 ± 0.46	80.39 ± 0.94	55.52 ± 0.84	65.32 ± 0.59	74.47 ± 0.60	85.87 ± 0.39
FedFomo [38]	65.02 ± 0.62	80.24 ± 0.97	65.16 ± 0.24	81.51 ± 0.54	56.72 ± 0.35	66.33 ± 0.74	75.17 ± 0.33	86.83 ± 0.24
FedPHP [39]	65.17 ± 0.49	81.10 ± 0.20	66.23 ± 0.61	82.42 ± 0.22	57.96 ± 0.73	67.35 ± 0.65	76.22 ± 0.54	87.88 ± 0.35
PMPFSL	$\mathbf{66.05 \pm 0.90}$	$\mathbf{82.35 \pm 0.55}$	$\mathbf{66.96 \pm 0.72}$	$\mathbf{83.53 \pm 0.36}$	$\mathbf{59.12 \pm 0.53}$	$\mathbf{68.20 \pm 0.38}$	$\mathbf{77.24 \pm 0.32}$	$\mathbf{88.66 \pm 0.37}$
UC Merced Dataset								
FedAvg [30]	46.39 ± 0.21	63.87 ± 0.58	42.31 ± 0.58	50.57 ± 0.11	39.04 ± 0.32	60.39 ± 0.54	58.03 ± 0.47	76.59 ± 0.18
FedProx [31]	47.09 ± 0.39	64.44 ± 0.67	43.01 ± 0.36	51.27 ± 0.55	40.01 ± 0.58	61.23 ± 0.60	58.92 ± 0.16	77.18 ± 0.25
FedProto [32]	47.89 ± 0.52	65.07 ± 0.35	43.73 ± 0.63	52.06 ± 0.27	40.53 ± 0.74	61.83 ± 0.67	59.72 ± 0.33	77.87 ± 0.42
PerFedAvg [36]	48.62 ± 0.63	66.00 ± 0.20	44.44 ± 0.42	52.84 ± 0.51	41.41 ± 0.27	62.50 ± 0.41	60.46 ± 0.36	78.37 ± 0.48
pFedMe [37]	49.33 ± 0.32	66.59 ± 0.14	45.41 ± 0.47	53.62 ± 0.31	42.06 ± 0.23	63.52 ± 0.42	61.37 ± 0.32	78.95 ± 0.41
FedFomo [38]	50.30 ± 0.81	67.31 ± 0.26	46.25 ± 0.82	54.53 ± 0.73	42.91 ± 0.75	64.28 ± 0.56	62.41 ± 0.20	79.93 ± 0.26
FedPHP [39]	51.02 ± 0.78	68.23 ± 0.28	47.14 ± 0.77	55.27 ± 0.33	43.78 ± 0.48	65.04 ± 0.15	63.19 ± 0.37	80.77 ± 0.67
PMPFSL	$\mathbf{51.96 \pm 0.42}$	$\mathbf{69.14 \pm 0.47}$	$\mathbf{48.06 \pm 0.37}$	$\mathbf{56.12 \pm 0.58}$	$\mathbf{44.75 \pm 0.47}$	$\mathbf{65.95 \pm 0.56}$	$\mathbf{64.08 \pm 0.35}$	$\mathbf{81.69 \pm 0.42}$
NWPU-RESISC45 Dataset								
FedAvg [30]	55.82 ± 0.48	73.04 ± 0.25	59.78 ± 0.63	68.24 ± 0.50	39.40 ± 0.12	57.33 ± 0.46	66.70 ± 0.27	84.51 ± 0.35
FedProx [31]	56.34 ± 0.44	73.64 ± 0.60	60.32 ± 0.42	68.72 ± 0.38	39.62 ± 0.54	57.62 ± 0.50	67.35 ± 0.52	85.08 ± 0.55
FedProto [32]	57.11 ± 0.27	73.57 ± 0.57	61.15 ± 0.23	69.35 ± 0.37	40.02 ± 0.29	58.04 ± 0.60	68.13 ± 0.30	85.57 ± 0.29
PerFedAvg [36]	58.01 ± 0.60	74.19 ± 0.60	61.77 ± 0.64	70.84 ± 0.29	40.14 ± 0.15	58.99 ± 0.44	68.85 ± 0.18	86.29 ± 0.12
pFedMe [37]	58.47 ± 0.58	74.72 ± 0.14	62.57 ± 0.49	72.76 ± 0.35	41.13 ± 0.51	59.77 ± 0.23	69.45 ± 0.43	87.23 ± 0.57
FedFomo [38]	59.06 ± 0.46	75.15 ± 0.51	63.02 ± 0.42	72.49 ± 0.27	41.00 ± 0.43	60.37 ± 0.46	70.28 ± 0.41	87.77 ± 0.56
FedPHP [39]	59.64 ± 0.63	75.24 ± 0.60	63.89 ± 0.51	73.57 ± 0.69	42.04 ± 0.36	61.27 ± 0.43	71.31 ± 0.15	88.40 ± 0.26
PMPFSL	$\mathbf{60.63 \pm 0.51}$	$\mathbf{76.56 \pm 0.29}$	$\mathbf{64.29 \pm 0.58}$	$\mathbf{74.75 \pm 0.53}$	$\mathbf{42.96 \pm 0.58}$	$\mathbf{62.16 \pm 0.25}$	$\mathbf{72.67 \pm 0.19}$	$\mathbf{89.54 \pm 0.17}$

respectively. For the experimental results of MPL, in addition to analyzing the performance of the participants' personalized models, we report the overall accuracy of the global model.

C. Compared With State-of-the-Art

1) Compared With Few-Shot Scene Classification SOTA: To demonstrate the effectiveness of our proposed method and ensure fair comparisons, we compared the PMPFSL framework with 21 few-shot peers on three benchmark FSSC datasets. The comparative methods include MatchingNet [45], model-agnostic meta-learning (MAML) [54], ProtoNet [46], transductive propagation network (TPN) [55], meta-stochastic gradient descent (Meta-SGD) [56], Relation Net [50], LLSR [12], RS-MetaNet [59], DLA-MatchNet [13], iterative distribution learning network (IDLN) [57], prototype calibration with feature generation network (PCFGNet) [58], metakernel networks (MKN) [60], self-supervised contrastive metric learning network (SCL-MLNet) [61], GES-Net [14], TAE-Net [44], SPNet [15], TSC [16], SGMNet [17], MLLD [20], MLCMNet [33], and MMML [21], where the subscripts in bold indicate that the corresponding studies were published in remote sensing journals. To be fair, their effects are reported in the best results of respective papers.

Tables I–III show the comparison results of PMPFSL and outstanding peers on the three benchmark datasets WHU-RS19, UC Merced, and NWPU-RESISC45, respectively. It is worth noting that "†" indicates that transductive inference is used, while others are inductive inference. It can be seen from the experimental results that our proposed MSFMNet obtains competitive results in the centralized machine learning paradigm. Although MSFMNet fails to rank first or second in some of the metrics, the convenience and portability of MSFMNet combined to obtain such a result is desirable compared to other FSSC algorithms with elaborate and complex designs. In particular, on the NWPU-RESISC45 dataset with large intra-class differences and inter-class similarities, MSFMNet can effectively distinguish samples between the support set and query set more discriminatively by constructing the self-attention correlations at multiscales.

In the MPL paradigm, MSFMNet likewise achieved the best results. Specifically, PMPFSL[MSFMNet] is at least 7.29% ahead of the second place in the MPL paradigm, whether in small-scale or large-scale few-shot remote sensing scene classification dataset. Since the training paradigm of MPL does not involve data, but only aggregation with multiparty trained models, it should theoretically be inferior to centralized machine learning in terms of few-shot classification performance. However, except on the WHU-RS19 dataset, different few-shot networks outperform centralized machine learning in MPL paradigm. This is largely due to the fact that the design of the PAD scheme allows for better generalization of the global aggregation model, rather than simple average aggregation.

Nevertheless, there are scenarios where the global model does not perform well, and Fig. 4 gives some examples of failed episodes from MSFMNet on the three benchmark few-shot datasets. Most of the cases of MSFMNet fail due to

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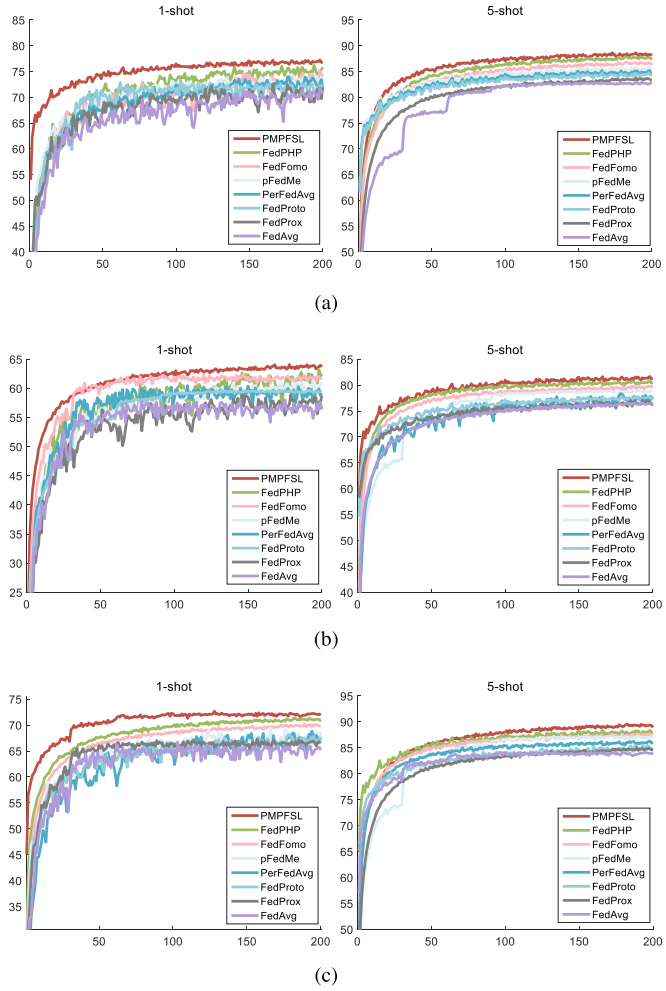


Fig. 5. Visualization results of the local models for PMPFSL on three benchmark datasets. (a) WHU-RS19 dataset. (b) UC Merced dataset. (c) NWPU-RESISC45 dataset.

the number of samples is too small, especially in the case of one-shot. It is attributed to the fact that each episode is randomly selected in the testing, and it is difficult for the model to distinguish between the labeled samples in the support set and the unlabeled samples in the query set when the overall scene is too far apart. In addition, the model also performs poorly when there are similar classes in the support set.

2) Compared With MPL and Personalized MPL SOTA: To validate the performance of our designed PMPFSL paradigm in multiparty few-shot remote sensing scene classification, we compared with MPL and personalized MPL algorithms FedAvg [30], FedProx [31], FedProto [32], PerFedAvg [36], pFedMe [37], FedFomo [38], and FedPHP [39], in four networks (ProtoNet, RelationNet, MatchingNet, and MSFMNet) as global and local models with the few-shot learning setting as the training paradigm. Detailed experimental results are given in Table IV.

For different MPL and personalized MPL algorithms, our designed PMPFSL achieves the best results on all four baseline networks. Specifically, on the large-scale NWPU-RESISC45 dataset, PMPFSL with our designed MSFMNet leads the second-place personalized MPL algorithm (FedPHP) by 1.36% in the one-shot setting and 1.14% in the five-shot setting. Moreover, the improvement of PMPFSL from one-shot to five-shot is also the most obvious, especially for the WHU-RS19 dataset with a small amount of data. Furthermore, we can observe that the personalized MPL algorithms clearly perform better on different few-shot networks compared to the normal MPL algorithms (FedAvg, FedProx, and FedProto), and the proposed PMPFSL is a standout.

Not limited to the overall performance of the global aggregation model, in Fig. 5, we give the visualization results of the local model accuracy and the number of communication rounds for one of the participants on these three benchmark datasets. It can be observed from Fig. 5 that PMPFSL achieves the best results in both convergence and local model accuracy on the one-shot and five-shot settings. This is because the PMPFSL framework we designed is not just to pursue a global model with outstanding capabilities, but also to design a personalized strategy to decouple the global model into the local personalized models that are more suitable for local data. From the comparison results, it can be concluded that PMPFSL not only performs well on the global aggregation model, but for different participants, PMSLE still gives a more personalized model that fits the local data.

D. Comprehensive Performance Analysis

1) Impact of the Number of Participants: Fig. 6 gives a comparison of the performance of different numbers of participants for PMPFSL on three benchmark datasets. The different numbers of participants imply that the number of local models that need to be processed in the global model aggregation phase is not the same, and it can be observed that as the number of participants increases the performance of the global aggregated models gradually decreases. This is largely attributed to the increased data heterogeneity, which makes the generalization ability of the global model decrease. However, for each participant's local model, the design of PAD in PMPFSL ensures that the local model is more adapted to the respective local data.

2) Impact of the Number of Local Update Rounds: Fig. 7 illustrates the effect of different numbers of local update rounds, also known as the parameter of episodes in few-shot learning, on the accuracy of the global aggregation model. It can be observed that as the number of rounds of local model update increases, the convergence of the global model is significant on small-scale scene classification datasets such as WHU-RS19 and UC Merced. Without loss of generality, convergence of the global model can be achieved after only a few tens of rounds or even less if the participants have certain computational resources to increase the number of local update rounds.

3) Experiments on the Personalized Adaptive Distillation Scheme: The PAD ablation experiments for PMPFSL are given in Table V, where "w/" and "w/o" represent the results with and without the PAD scheme in the baseline PMPFSL, respectively. It can be concluded that in all four of these few-shot networks, the design of the PAD resulted in a substantial improvement in the baseline. Moreover, the improvement of PAD is universal and not limited to a certain few-shot network, but applies to a wide variety

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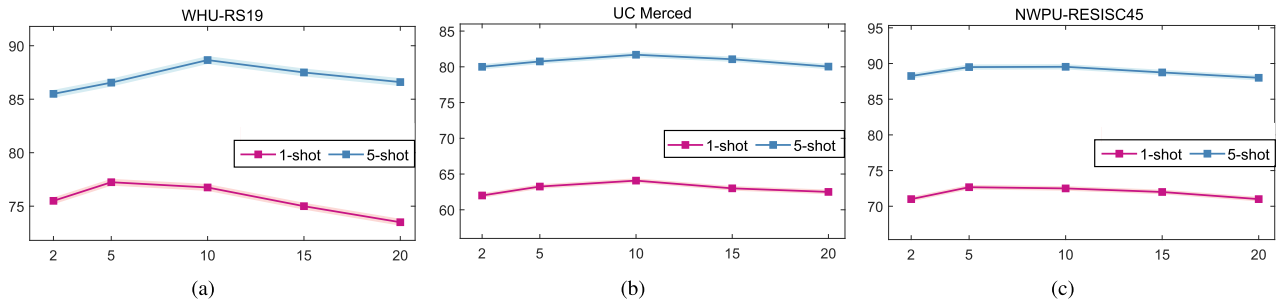


Fig. 6. Performance comparison of PMPFSL with different numbers of participants. (a) WHU-RS19 dataset. (b) UC Merced dataset. (c) NWPU-RESISC45 dataset.

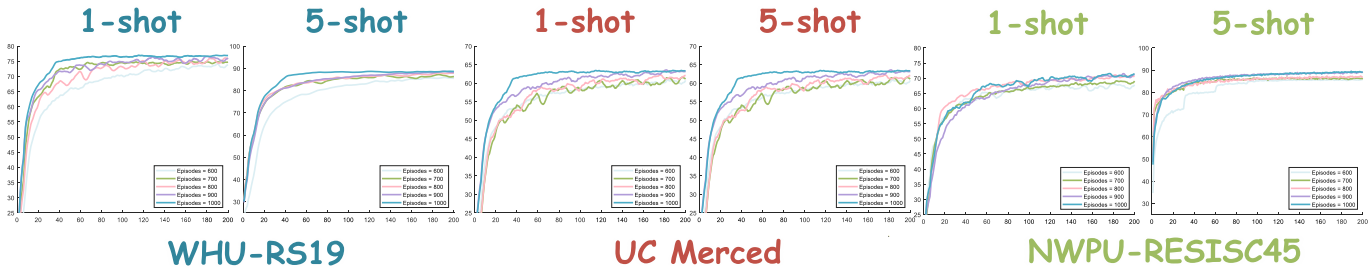


Fig. 7. Performance comparison of PMPFSL with different numbers of local update episodes.

TABLE V

ABLATION RESULTS (IN %) OF PAD ON THREE BENCHMARK DATASETS FOR THE FOUR BASELINE NETWORKS

Dataset	Method	**PMPFSL**[ProtoNet]		**PMPFSL**[RelationNet]		**PMPFSL**[MatchingNet]		**PMPFSL**[**MSFMNet**]	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
WHU-RS19	w/o PAD	61.54 ± 0.47	75.71 ± 0.50	62.07 ± 0.35	77.11 ± 0.88	52.83 ± 0.42	62.89 ± 0.19	71.90 ± 0.34	82.94 ± 0.61
	w/ PAD	$\mathbf{66.05 \pm 0.90}$	$\mathbf{82.35 \pm 0.55}$	$\mathbf{66.96 \pm 0.72}$	$\mathbf{83.53 \pm 0.36}$	$\mathbf{59.12 \pm 0.53}$	$\mathbf{68.20 \pm 0.38}$	$\mathbf{77.24 \pm 0.32}$	$\mathbf{88.66 \pm 0.37}$
UCM	w/o PAD	46.39 ± 0.21	63.87 ± 0.58	42.31 ± 0.58	50.57 ± 0.11	39.04 ± 0.32	60.39 ± 0.54	58.03 ± 0.47	76.59 ± 0.18
	w/ PAD	$\mathbf{51.96 \pm 0.42}$	$\mathbf{69.14 \pm 0.47}$	$\mathbf{48.06 \pm 0.37}$	$\mathbf{56.12 \pm 0.58}$	$\mathbf{44.75 \pm 0.47}$	$\mathbf{65.95 \pm 0.56}$	$\mathbf{64.08 \pm 0.35}$	$\mathbf{81.69 \pm 0.42}$
NWPU45	w/o PAD	55.82 ± 0.48	73.04 ± 0.25	59.78 ± 0.63	68.24 ± 0.50	39.40 ± 0.12	57.33 ± 0.46	66.70 ± 0.27	84.51 ± 0.35
	w/ PAD	$\mathbf{60.63 \pm 0.51}$	$\mathbf{76.56 \pm 0.29}$	$\mathbf{64.29 \pm 0.58}$	$\mathbf{74.75 \pm 0.53}$	$\mathbf{42.96 \pm 0.58}$	$\mathbf{62.16 \pm 0.25}$	$\mathbf{72.67 \pm 0.19}$	$\mathbf{89.54 \pm 0.17}$

of well-designed few-shot networks, which further demonstrates the effectiveness and superiority of PAD in FSSC task for different personalized local models. In addition, Table IV also reflects that the design of PAD also achieves the best results compared to other personalized MPL algorithms (PerFedAvg, pFedMe, FedFomo, and FedPHP) as well.

4) Experiments on the MSFMNet: In Table IV, MSFMNet as a few-shot baseline network achieves the best results on both different MPL and personalized MPL strategies. It illustrates that compared with other few-shot networks with a single metric, the design of the multiscale feature metric matching strategy in MSFMNet is more capable of fully exploring the correlation between the support set and the query set, and thus obtaining better few-shot classification performance. To further illustrate the superiority of MSFMNet, Fig. 8 gives a comparison of the heatmaps of MSFMNet and other few-shot networks on some representative test cases. Specifically, in the case of "beach," other few-shot networks can only correlate the splash and beach parts, while MSFMNet is able to focus on the coastline as a way of judging it as a beach scenario. In addition, a comparison of the t-SNE visualization of MSFMNet and other few-shot networks is given in Fig. 9. It can be observed that for large-scale data, neither MatchingNet, ProtoNet, nor Relation Net is able to make a more discriminative judgment, especially for the samples with high inter-class similarity. MSFMNet, on the other hand, is able to distinguish each class of complex scenarios well, which provides strong support for the subsequent metrics.

5) Experiments on the Time Cost: Table VI gives the results of comparing the time cost of different MPL algorithms based on the MSFMNet. It can be seen from the experimental results that the time for MPL methods is generally related to the size of the datasets, which is usually due to the fact that local model updates naturally take longer if the data is large. More specifically, the classical average aggregation algorithm, FedAvg, obtains the least time cost, which results from the fact that it only requires local model updates and global aggregation. Whereas among the personalized MPL algorithms, our proposed PMPFSL achieves the best compromise between time cost and classification accuracy. Moreover, Table VII

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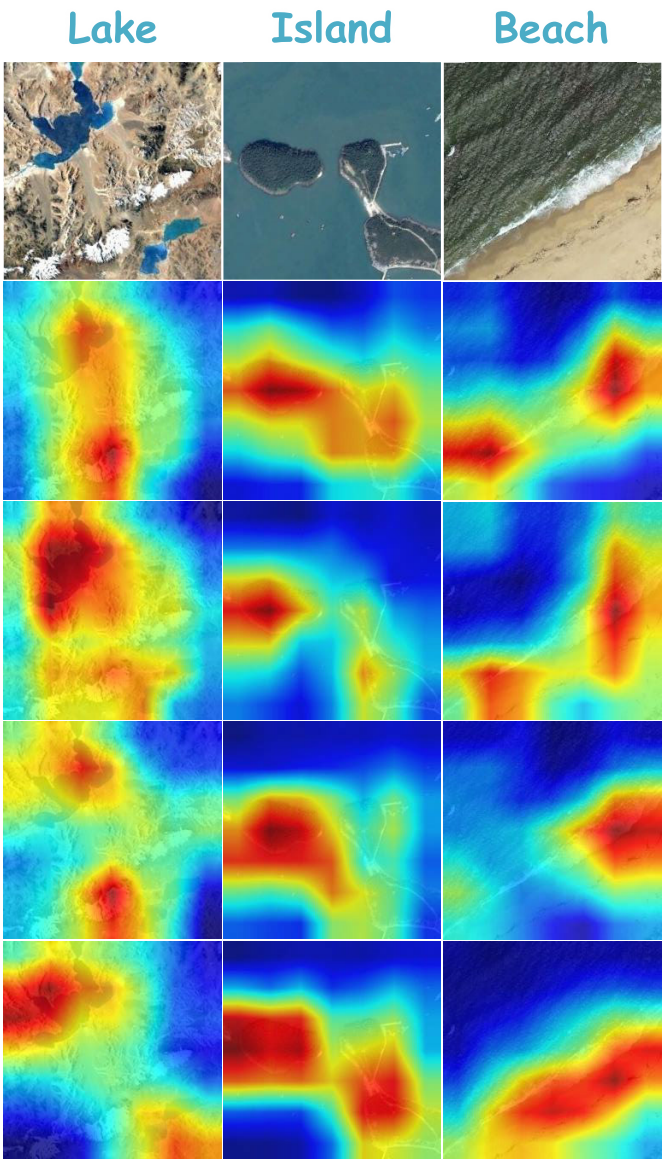


Fig. 8. Comparison of heatmaps for some representative cases. The rows from top to bottom correspond to the original image, heatmaps obtained by MatchingNet, ProtoNet, Relation Net, and MSFMNet.

TABLE VI

COMPARATIVE RESULTS (IN GPU DAY) OF THE TIME COST FOR DIFFERENT MPL METHODS ON MSMNET. NOTE: THE "GPU DAY" REFERS TO THE TIME BASED ON A SINGLE NVIDIA GEFORCE RTX 4090 GRAPHICS CARD

Method	WHU-RS19		UC Merced		NWPU-RESISC45	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
FedAvg [30]	**0.0347**	**0.0403**	**0.0583**	**0.0764**	**0.1014**	**0.1431**
FedProx [31]	0.0479	0.0590	0.0868	0.1125	0.1458	0.2083
FedProto [32]	0.1215	0.1500	0.2069	0.2958	0.3125	0.3639
PerFedAvg [36]	0.0639	0.0785	0.1111	0.1507	0.2014	0.2604
pFedMe [37]	0.0972	0.1215	0.1694	0.2257	0.3042	0.3299
FedFomo [38]	0.0563	0.0681	0.1007	0.1236	0.1701	0.2340
FedPHP [39]	0.1028	0.1285	0.1847	0.2153	0.2500	0.2875
PMPFSL	0.0451	0.0556	0.0806	0.1042	0.1354	0.1875

gives the time cost of four few-shot networks in the centralized machine learning paradigm. The simpler metrics ProtoNet with cosine distance and MatchingNet with Euclidean distance consume less time, but sacrifice a degree of classification accuracy.

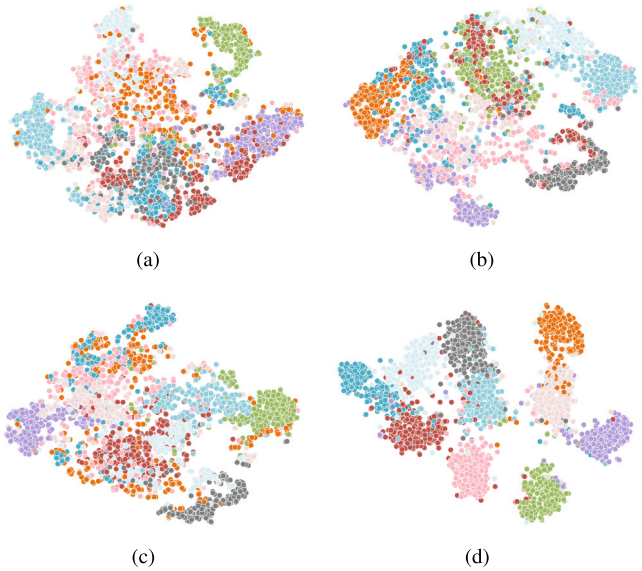


Fig. 9. t-SNE visualization comparison results on the NWPU-RESISC45 dataset. (a) MatchingNet. (b) ProtoNet. (c) Relation Net. (d) MSFMNet.

TABLE VII

COMPARATIVE RESULTS (IN GPU DAY) OF THE TIME COST FOR DIFFERENT FEW-SHOT NETWORKS IN THE CENTRALIZED MACHINE LEARNING PARADIGM

Method	WHU-RS19		UC Merced		NWPU-RESISC45	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
ProtoNet	**0.0153**	**0.0271**	**0.0188**	**0.0313**	**0.0240**	**0.0375**
RelationNet	0.0188	0.0299	0.0229	0.0403	0.0278	0.0521
MatchingNet	0.0167	0.0275	0.0194	0.0319	0.0243	0.0382
MSFMNet	0.0215	0.0313	0.0278	0.0451	0.0368	0.0597

V. CONCLUSION

In this article, we proposed a groundbreaking PMPFSL paradigm, tailored specifically for remote sensing scene classification. PMPFSL serves as a cutting-edge baseline for multiparty collaborative learning, seamlessly integrating with diverse MPL algorithms and few-shot classification networks. Drawing upon this paradigm, we propose a PAD scheme and an MSFMNet, tailored, respectively, to multiparty collaborative learning and few-shot network in remote sensing scene classification. Specifically, PAD discards the traditional approach of relying solely on local iterations for personalization of the global model. Instead, it adaptively selects the shallow features from the global model that are pertinent to local data, personalizes the deep features, and mitigates the overfitting issue in the local model with knowledge distillation. On the other hand, MSFMNet focuses on extracting self-attention correlation features from support and query sets at multiple scales, enabling more efficient metrics with a limited number of labeled samples. Overall, PMPFSL revolutionizes the training paradigm for few-shot remote sensing scene classification. It enables participants to engage in collaborative training without disclosing their

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private data, resulting in a more generalized and personalized model.

In the future, we will focus on extending PMPFSL to more few-shot remote sensing tasks, such as target detection, object segmentation, and realizing personalized multiparty collaborative few-shot learning for full-scene multitasks in remote sensing.

REFERENCES

[1] A. Ma, Y. Wan, Y. Zhong, J. Wang, and L. Zhang, "SceneNet: Remote sensing scene classification deep learning network using multi-objective neural evolution architecture search," *ISPRS J. Photogramm. Remote Sens.*, vol. 172, pp. 171–188, Feb. 2021.

[2] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[3] K. Nogueira, O. A. B. Penatti, and J. A. Dos Santos, "Towards better exploiting convolutional neural networks for remote sensing scene classification," *Pattern Recognit.*, vol. 61, pp. 539–556, Jan. 2017.

[4] R. M. Anwer, F. S. Khan, J. van de Weijer, M. Molinier, and J. Laaksonen, "Binary patterns encoded convolutional neural networks for texture recognition and remote sensing scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 74–85, Apr. 2018.

[5] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[6] J. Zhu, K. Yang, N. Guan, X. Yi, and C. Qiu, "HCPNet: Learning discriminative prototypes for few-shot remote sensing image scene classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 123, Sep. 2023, Art. no. 103447.

[7] N. He, L. Fang, S. Li, J. Plaza, and A. Plaza, "Skip-connected covariance network for remote sensing scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1461–1474, May 2020.

[8] W.-Z. Li, J.-W. Zhou, X. Li, Y. Cao, and G. Jin, "Few-shot object detection on aerial imagery via deep metric learning and knowledge inheritance," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103397.

[9] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[10] W. Xu, J. Wang, Z. Wei, M. Peng, and Y. Wu, "Deep semantic-visual alignment for zero-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 140–152, Apr. 2023.

[11] M. Gong, J. Li, Y. Zhang, Y. Wu, and M. Zhang, "Two-path aggregation attention network with quad-patch data augmentation for few-shot scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4511616.

[12] M. Zhai, H. Liu, and F. Sun, "Lifelong learning for scene recognition in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1472–1476, Sep. 2019.

[13] L. Li, J. Han, X. Yao, G. Cheng, and L. Guo, "DLA-MatchNet for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7844–7853, Sep. 2021.

[14] Z. Yuan, W. Huang, C. Tang, A. Yang, and X. Luo, "Graph-based embedding smoothing network for few-shot scene classification of remote sensing images," *Remote Sens.*, vol. 14, no. 5, p. 1161, Feb. 2022.

[15] G. Cheng et al., "SPNet: Siamese-prototype network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5608011.

[16] Q. Zeng and J. Geng, "Task-specific contrastive learning for few-shot remote sensing image scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 143–154, Sep. 2022.

[17] B. Zhang et al., "SGMNet: Scene graph matching network for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628915.

[18] Y. Ma, J. Meng, B. Liu, L. Sun, H. Zhang, and P. Ren, "Dictionary learning for few-shot remote sensing scene classification," *Remote Sens.*, vol. 15, no. 3, p. 773, Jan. 2023.

[19] Y. Jia, J. Gao, W. Huang, Y. Yuan, and Q. Wang, "Exploring hard samples in multiview for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615714.

[20] Z. Yuan, C. Tang, A. Yang, W. Huang, and W. Chen, "Few-shot remote sensing image scene classification based on metric learning and local descriptors," *Remote Sens.*, vol. 15, no. 3, p. 831, Feb. 2023.

[21] X. Chen, G. Zhu, and J. Wei, "MMML: Multimanifold metric learning for few-shot remote-sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5618714.

[22] Y. Xu, T. Bai, W. Yu, S. Chang, P. M. Atkinson, and P. Ghamisi, "AI security for geoscience and remote sensing: Challenges and future trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 2, pp. 60–85, Jun. 2023.

[23] J. Li et al., "MSANet: Multiscale self-attention aggregation network for few-shot aerial imagery segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4500216.

[24] K. Wei et al., "Federated learning with differential privacy: Algorithms and performance analysis," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 3454–3469, 2020.

[25] Y. Sun, L. Shen, H. Sun, L. Ding, and D. Tao, "Efficient federated learning via local adaptive amended optimizer with linear speedup," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14453–14464, Dec. 2023.

[26] F. Sattler, S. Wiedemann, K.-R. Müller, and W. Samek, "Robust and communication-efficient federated learning from non-i.I.d. data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 9, pp. 3400–3413, Sep. 2020.

[27] X. Zhang, B. Zhang, W. Yu, and X. Kang, "Federated deep learning with prototype matching for object extraction from very-high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603316.

[28] Y. Song and G. Dong, "Federated target recognition for multiradar sensor data security," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5207310.

[29] W. Cai, M. Gao, Y. Ding, X. Ning, X. Bai, and P. Qian, "Stereo attention cross-decoupling fusion-guided federated neural learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527516.

[30] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist.*, vol. 54. Fort Lauderdale, FL, USA, 2017, pp. 1273–1282.

[31] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst.*, vol. 2, 2020, pp. 429–450.

[32] Y. Tan et al., "FedProto: Federated prototype learning across heterogeneous clients," in *Proc. AAAI*, 2022, pp. 8432–8440.

[33] X. Chen, G. Zhu, M. Liu, and Z. Chen, "Few-shot remote sensing image scene classification based on multiscale covariance metric network (MCMNet)," *Neural Netw.*, vol. 163, pp. 132–145, Jun. 2023.

[34] H. Zhu, X. Wang, and Y. Jin, "Federated many-task Bayesian optimization," *IEEE Trans. Evol. Comput.*, doi: 10.1109/TEVC.2023.3279775.

[35] J. Chen, J. Xue, Y. Wang, L. Huang, T. Baker, and Z. Zhou, "Privacy-preserving and traceable federated learning for data sharing in industrial IoT applications," *Expert Syst. Appl.*, vol. 213, Mar. 2023, Art. no. 119036.

[36] A. Fallah, A. Mokhtari, and A. Ozdaglar, "Personalized federated learning: A meta-learning approach," 2020, *arXiv:2002.07948*.

[37] C. T. Dinh, N. Tran, and T. D. Nguyen, "Personalized federated learning with Moreau envelopes," in *Proc. NeurIPS*, vol. 33, 2020, pp. 21394–21405.

[38] M. Zhang, K. Sapra, S. Fidler, S. Yeung, and J. M. Alvarez, "Personalized federated learning with first order model optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–11.

[39] X.-C. Li, D.-C. Zhan, Y. Shao, B. Li, and S. Song, "FedPHP: Federated personalization with inherited private models," in *Proc. Eur. Conf. ECML PKDD*, Bilbao, Spain. Cham, Switzerland: Springer, Sep. 2021, pp. 587–602.

[40] J. Zhu, J. Wu, A. K. Bashir, Q. Pan, and W. Yang, "Privacy-preserving federated learning of remote sensing image classification with dishonest majority," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4685–4698, 2023.

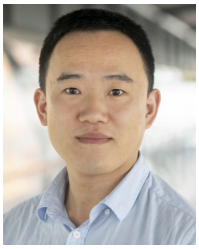
[41] J. Li et al., "Toward multiparty personalized collaborative learning in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4503616.

[42] J. Li, M. Gong, H. Liu, Y. Zhang, M. Zhang, and Y. Wu, "Multiform ensemble self-supervised learning for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4500416.

[43] Y. Zhang, M. Gong, M. Zhang, and J. Li, "Self-supervised monocular depth estimation with self-perceptual anomaly handling," *IEEE Trans. Neural Netw. Learn. Syst.*, doi: 10.1109/TNNLS.2023.3301711.

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[44] W. Huang, Z. Yuan, A. Yang, C. Tang, and X. Luo, "TAE-Net: Task-adaptive embedding network for few-shot remote sensing scene classification," *Remote Sens.*, vol. 14, no. 1, p. 111, Dec. 2021.
[45] O. Vinyals, C. Blundell, T. Lillicrap, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2016, pp. 3630–3638.
[46] J. Snell, K. Swersky, and R. S. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 30, 2017, pp. 4077–4087.
[47] J. Soria-Comas, J. Domingo-Ferrer, D. Sánchez, and D. Megías, "Individual differential privacy: A utility-preserving formulation of differential privacy guarantees," *IEEE Trans. Inf. Forensics Security*, vol. 12, pp. 1418–1429, 2017.
[48] X. Yan, G. Zhou, Y. Huang, W. Meng, A.-T. Nguyen, and H. Huang, "Secure estimation using partially homomorphic encryption for unmanned aerial systems in the presence of eavesdroppers," *IEEE Trans. Intell. Veh.*, doi: 10.1109/TIV.2024.3378288.
[49] X. Li, J. Niu, M. Z. A. Bhuiyan, F. Wu, M. Karuppiah, and S. Kumari, "A robust ECC-based provable secure authentication protocol with privacy preserving for industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 14, no. 8, pp. 3599–3609, Aug. 2018.
[50] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. S. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1199–1208.
[51] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[52] G. Sheng, W. Yang, T. Xu, and H. Sun, "High-resolution satellite scene classification using a sparse coding based multiple feature combination," *Int. J. Remote Sens.*, vol. 33, no. 8, pp. 2395–2412, Apr. 2012.
[53] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.
[54] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, vol. 70, Aug. 2017, pp. 1126–1135.
[55] Y. Liu, "Learning to propagate labels: Transductive propagation network for few-shot learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018, pp. 1–14.
[56] Z. Li, F. Zhou, F. Chen, and H. Li, "Meta-SGD: Learning to learn quickly for few-shot learning," 2017, *arXiv:1707.09835*.
[57] Q. Zeng, J. Geng, W. Jiang, K. Huang, and Z. Wang, "IDLN: Iterative distribution learning network for few-shot remote sensing image scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[58] Q. Zeng, J. Geng, K. Huang, W. Jiang, and J. Guo, "Prototype calibration with feature generation for few-shot remote sensing image scene classification," *Remote Sens.*, vol. 13, no. 14, p. 2728, Jul. 2021.
[59] H. Li et al., "RS-MetaNet: Deep metametric learning for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6983–6994, Aug. 2021.
[60] Z. Cui, W. Yang, L. Chen, and H. Li, "MKN: Metakernel networks for few shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4705611.
[61] X. Li, D. Shi, X. Diao, and H. Xu, "SCL-MLNet: Boosting few-shot remote sensing scene classification via self-supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5801112.



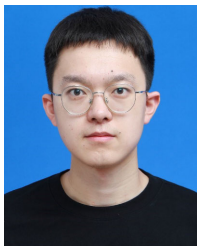
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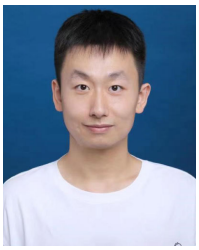
Dr. Gong is the Executive Committee Member of the Chinese Association for Artificial Intelligence and a Senior Member of the Chinese Computer Federation. He was a recipient of the prestigious National Program for Support of the Leading Innovative Talents from the Central Organization Department of China, the Leading Innovative Talent in Science and Technology from the Ministry of Science and Technology of China, the Excellent Young Scientist Foundation from the National Natural Science Foundation of China, the New Century Excellent Talent from the Ministry of Education of China, and the National Natural Science Award of China. He is an Associate Editor or an Editorial Board Member for over five journals, including IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.

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